Sai Akhila Reddy Turpu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.75
Narayana Jr. College, Nallakunta	Class XII (SSC)	2024	97.40%
Johnson Grammar School	Class X (ICSE)	2022	98.40%

EXPERIENCE

Texas Instruments WiSH Embedded Systems Mentorship Program:

- Completed hands-on training in embedded software, ARM architecture, and real-time embedded systems.
- Developed Embedded C applications on the **TI C2000 LaunchPad** and **ControlCARD** using **Code Composer Studio (CCS)**.
- Implemented hardware interfaces including LED control, IR sensing, **ultrasonic** distance measurement **(eCAP)**, **PWM-based stepper motor control**, and UART communication.
- Gained a solid understanding of **ARM Cortex-M3, Embedded C, pointers, cross-compilation, Makefiles, Git and communication protocols.**

PROJECTS

Inter-IIT Tech Meet 14.0 ETERNAL PS | Github

- **Designing ROS2 Architecture:** Implemented kinematics and odometry estimation nodes in Raspberry Pi for a 4-wheel mecanum-drive.
- **Navigation:** Interacted with **Nav2** and SLAM-ToolBox stack for real-time mapping and path planning. Developed a custom **OpenCV-based keepout filter** with automated rack detection, overlaid it on the Nav2 costmap layer, and enforced no-go areas.
- **Hardware integration:** Implemented Raspberry Pi-ESP32 communication for velocity command exchange. Implemented joystick teleop node for manual control.
- **Simulation:** Simulated Navigation stack in Gazebo for verifying the accuracy of keep-out filter.

WiSH Design Thinking(Texas Instruments) | Github

- Designed a **solar-powered IoT water quality monitoring system** for rural villages using the **Design Thinking** approach.
- Architected an end-to-end solution integrating **IoT sensors, LoRa/4G, cloud analytics, and SMS alerts** for real-time monitoring.
- Developed a prototype with a **TDS sensor** and a dashboard for real-time water quality visualization.
- Designed a **low-cost (~10K/node)**, scalable architecture for contamination detection and pipeline fault localization.
- Defined user personas, success metrics, and alert workflows to improve rural water quality monitoring and management.

Biometric Attendance System| Cloud | Github

- Developed a **multi-modal biometric attendance system** in Python integrating **facial recognition and fingerprint authentication** with configurable edge-device deployment.
- Designed a **local-first architecture** using SQLite for offline attendance logging, with automatic synchronization to **cloud MySQL and REST-based HR/ERP APIs**.
- Implemented **parallel biometric processing, background employee synchronization, and fault-tolerant data queuing** for reliable real-time attendance tracking.

Sampling theorem and Reconstruction | Github

- Designed and implemented an **8-bit SAR ADC** with a **Sample-and-Hold** circuit using **Arduino, a comparator, and MCP4921 DACs**, with firmware for SAR conversion and SPI-based DAC control.
- Experimentally verified the **Nyquist-Shannon Sampling Theorem** by demonstrating **aliasing and Nyquist-rate,** and reconstructed analog signals using an **RC low-pass filter**.
- Evaluated ADC performance through **resolution, gain error, differential nonlinearity (DNL), and integral nonlinearity (INL)** measurements.

SKILLS

Programming languages: C, Embedded C, Python, Assembly in ARM Cortex m3*, Java*

Tools: LTSpice, Altium Designer, MATLAB, Git, Github, Arduino IDE, CCS(Code Composer Studio) IDE, Linux, Intel Quartus Lite

Frameworks: Numpy, Matplotlib, OpenCV, ROS2, Nav2, SLAM Toolbox

RELEVANT COURSES

Mathematics: Math for EE, Vector Calculus, Matrix Theory, Differential Equations and Transform Techniques, Probability, Random Processes, Intro to Stats, Machine Learning

Electrical: Semiconductor Device Fundamentals, Analog Electronics, Scientific Programming for EE, Digital Systems, Signal Processing, PCB Design, Control Systems

Programming: Intro to Programming, DSA, Machine Learning

Labs: Electric circuits Lab, Electronic Devices lab, Digital Systems Lab, DSP Lab

POSITIONS OF RESPONSIBILITY

Core member - Robotix club, IIT Hyderabad

Coordinator - Elan & nVision, IIT Hyderabad.

Core - Gesture, the art club of IIT Hyderabad